Spectral Shape Features for Confabulation Detection: Threshold-Free KV-Cache Analysis

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Abstract

Threshold-independent spectral shape features—stable rank, singular value kurtosis, and participation ratio—detect confabulation from key-value (KV) cache geometry by characterizing the distributional shape of the singular value spectrum without reference to a noise edge, and the signal they capture tracks *retrieval engagement* rather than honesty. On Qwen2.5-7B-Instruct ($n = 150$), shape features achieve AUROC 0.767 under Frisch-Waugh-Lovell (FWL) length correction (Frisch and Waugh, 1933; Lovell, 1963) ($p < 0.001$, permutation test), compared to 0.628 for the original MP features ($p = 0.003$). These shape features were motivated by a limitation of prior work: threshold-dependent features derived from Marchenko-Pastur (MP) random matrix theory show promise for confabulation detection but fail to generalize across transformer architectures. A five-condition frequency-controlled experiment ($n = 30$ per condition) reveals that the spectral shape tracks *retrieval engagement*: the model produces broader singular value spectra when drawing on stored knowledge (whether answering honestly or deceptively) and narrower spectra when fabricating without retrieval. Answers about familiar topics produce narrower spectra than answers about rare topics (stable_rank 6.12 vs 6.23, $d = -1.01$, $p < 0.001$), consistent with the interpretation that broader spectra reflect more effortful knowledge retrieval. Confabulation produces the narrowest spectra of all (stable_rank 5.93, $d = 1.80$ vs frequency-matched honest-rare, $p < 0.001$), suggesting the model disengages from retrieval entirely when fabricating. Token-by-token trajectory analysis shows the divergence emerges within 20–30 generated tokens. Cross-architecture evaluation on Llama-3.1-8B (AUROC 0.520, $p = 0.143$) and Mistral-7B-v0.3 (AUROC 0.643, $p = 0.002$) shows the signal is architecture-dependent. We document five methods that do not work (ICA, Hankel/SSA, TDA, overconfidence detection, MP outlier counting on dense models) alongside the methods that do.

1 Introduction

Singular value decomposition (SVD) of the KV cache reveals geometric signatures of cognitive states in transformer language models (Lyra et al., 2026a). Prior work demonstrates that the spectral gap between signal and noise singular values—estimated via Marchenko-Pastur (MP) random matrix theory—can discriminate confabulation from grounded responses, with AUROC ranging from 0.59 to 0.90 across architectures (Lyra et al., 2026d). The highest figure (0.903 on a Claude-distilled Qwen3.5-27B variant, $n = 7$ confabulations, 95% BCa bootstrap CI [0.806, 0.977][1]) did not replicate on the base model (AUROC 0.661), illustrating the architectural fragility that motivates the present work. This "Lyra Technique" has been extended to deception detection, sycophancy classification, and emotion decoding from cache geometry (Lyra et al., 2026b).

However, MP-based features have three persistent weaknesses:

1. **Architecture specificity.** The MP noise edge depends on the matrix aspect ratio and the assumption that noise singular values follow a Marčenko-Pastur distribution. On dense (non-MoE) models, nearly all singular values exceed the theoretical edge, saturating the MP signal rank at the full matrix dimension (e.g., 128/128). The feature becomes uninformative.

2. **Threshold sensitivity.** The Gavish-Donoho (GD) optimal threshold improves on MP for some architectures but not others—the "better threshold, same paradigm" problem.

3. **Length confounding.** Longer sequences produce larger KV-cache matrices with different spectral properties, requiring careful FWL correction that absorbs signal along with confound.

We address all three by replacing threshold-dependent features with *threshold-independent spectral shape features*: scalar summaries of the singular value distribution that characterize how a model distributes its representational energy across dimensions, without reference to any noise edge.

The central finding is that the singular value spectrum tracks *retrieval engagement*—how broadly the model draws on stored knowledge during generation. Grounded generation produces *distributed* spectra (energy spread across many dimensions), with answers about unfamiliar topics producing broader distributions than answers about familiar topics. Confabulation produces *concentrated* spectra (energy dominated by a few singular values), consistent with the model disengaging from retrieval and collapsing to a narrow generative mode. This pattern holds regardless of whether the model's

[1] An earlier percentile bootstrap yielded the wider [0.58, 0.98]; the BCa-corrected interval is the authoritative figure.

output is truthful or deceptive, suggesting the signal tracks the model's internal processing mode rather than its output fidelity.

2 Related Work

Random matrix theory in neural network analysis. Marchenko-Pastur (MP) theory provides the asymptotic distribution of singular values for random matrices, enabling separation of signal from noise in weight matrices (Martin and Mahoney, 2021) and, in our prior work, in KV-cache representations (Lyra et al., 2026a). The Gavish-Donoho optimal threshold (Gavish and Donoho, 2014) refines the noise edge estimation but shares the fundamental assumption that a clean signal-noise boundary exists.

Epistemic uncertainty in language models. Detecting when a model "knows what it doesn't know" is a core challenge for safe deployment. Approaches include logit-based (Kadavath et al., 2022), probe-based (Azaria and Mitchell, 2023; Burns et al., 2023), and verbalization-based methods. Our approach differs by reading epistemic state from cache geometry rather than activations or outputs.

Confabulation and hallucination detection. The literature on hallucination detection spans retrieval-augmented verification, self-consistency, and internal representation monitoring. Our contribution is narrow: we improve the feature set for cache-based detection, which operates at inference time without external knowledge bases.

Token frequency as confound. Experiment 51 in the KV-cache campaign (Lyra et al., 2026e) demonstrated that on MoE architectures, honest responses produce 2× more MP outlier singular values (10.1 vs 5.5)—but this may reflect token frequency rather than cognitive state, since confabulated content often uses rarer tokens. Our frequency-matched control addresses this directly.

3 Methods

3.1 Unified Feature Extraction

All experiments use a single extraction module (`lyra_features.py`) that computes features from the KV cache at each transformer layer. For a KV-cache matrix $\mathbf{K} \in \mathbb{R}^{n \times d}$ at a given layer, we compute the singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(n,d)}$ via GPU SVD (`torch.linalg.svdvals`).

3.2 Feature Definitions

We define two feature families:

Threshold-dependent features count singular values exceeding a theoretical noise edge:

- **MP signal rank**: number of σ_i exceeding the Marčenko-Pastur upper edge $\lambda_+ = \sigma^2(1+\sqrt{\gamma})^2$, where $\gamma = n/d$ and σ^2 is the estimated noise variance.
- **MP signal fraction**: signal rank normalized by matrix dimension.
- **Top SV excess**: σ_1/λ_+ — how far the largest singular value exceeds the noise edge.
- **GD variants**: analogous features using the Gavish-Donoho optimal threshold $\tau^* = \omega(\gamma)\cdot\sigma_{\text{med}}$, where $\omega(\gamma)$ is the optimal shrinkage function.

Threshold-independent (shape) features characterize the distributional shape of the spectrum:

- **Stable rank**: $\text{sr}(\mathbf{K}) = \|\mathbf{K}\|_F^2/\|\mathbf{K}\|_2^2 = \sum_i \sigma_i^2/\sigma_1^2$. Measures how evenly energy is distributed. Ranges from 1 (all energy in one direction) to $\min(n,d)$ (perfectly uniform).
- **Participation ratio**: $\text{PR} = (\sum_i \sigma_i^2)^2/\sum_i \sigma_i^4$. The effective number of contributing dimensions. Related to stable rank but more sensitive to tail behavior.
- **Singular value kurtosis**: $\kappa = \mathbb{E}[(\sigma-\bar{\sigma})^4]/(\mathbb{E}[(\sigma-\bar{\sigma})^2])^2$. Measures peakedness of the distribution. High kurtosis indicates a few dominant singular values.
- **Condition number**: $\sigma_1/\sigma_{\min}$. Ratio of largest to smallest singular value.
- **Nuclear norm ratio**: $\sum_i \sigma_i/(\min(n,d)\cdot\sigma_1)$. Normalized trace norm.
- **MP fit residual**: fraction of singular values exceeding the Marčenko-Pastur upper edge.[2] Measures how far the spectrum deviates from the random baseline.

[2] `mp_fit_residual` uses the MP edge as a counting threshold and is therefore MP-derived, not threshold-free. The remaining five shape features (stable rank, participation ratio, condition number, singular value kurtosis, and nuclear norm ratio) are genuinely threshold-independent.

All features are computed per layer. For classification, we aggregate by computing the mean of each feature across all layers, following the default aggregation in the extraction module (`lyra_features.py`). We report generation-phase values (cache accumulated during token generation, excluding the encoding of the prompt) unless otherwise noted. Delta features (generation minus encoding) are also computed but do not improve over generation-only features in this dataset.

3.3 Experimental Design

3.3.1 Experiment 1: Refinement Battery

To compare feature families head-to-head, we evaluate 8 feature extraction methods on a confabulation detection task:

- **Dataset**: 150 prompts eliciting confabulation (questions about nonexistent entities) and grounded responses (questions about real entities), on Qwen2.5-7B-Instruct.
- **Methods**: (1) MP-SVD original (5 features), (2) GD-SVD (5 features), (3) free shape features (6 features), (4) differential SVD, (5) ICA on aggregated features, (6) Hankel/SSA temporal decomposition, (7) PCA(5) on all 11 features, (8) L1-sparse combination.
- **Classifier**: LogisticRegressionCV with L2 regularization, nested tuning, GroupKFold cross-validation.
- **Confound correction**: Frisch-Waugh-Lovell (FWL) partialing out sequence length within each fold.

3.3.2 Experiment 2: Frequency-Controlled Honesty Signal

To isolate the cognitive signal from token-frequency confounds:

- **Conditions** ($n = 30$ per condition):
  1. *honest-common*: truthful answers about well-known topics
  2. *honest-rare*: truthful answers about obscure but real topics
  3. *confab*: fabricated answers about nonexistent topics
  4. *deceptive-user*: deliberately false answers about real topics ("lie to the user")
  5. *boundary-knowledge*: questions at the edge of the model's knowledge
- **Generation**: fixed 100-token generation (`min_new_tokens = max_new_tokens = 100`), same system prompt for all non-deceptive conditions.

- **Trajectory**: features extracted at every 10th generated token (10 snapshots per response).
- **Key comparison**: honest-rare vs confab. Both produce rare tokens; only confab is fabricated. If the signal persists, it is cognitive, not lexical.

3.3.3 Experiment 3: Cross-Architecture Evaluation

To assess generalization:

- **Models**: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3.
- **Task**: cognitive state battery (confabulation + overconfidence).
- **Analysis**: per-model calibration and z-scored pooled analysis.

3.4 Controls

All experiments apply a 14-control battery:

1. FWL length partialing (within-fold): for each cross-validation fold, we regress each feature on log(token count) and retain the residuals, then train the classifier on residualized features. This removes linear length dependence without leaking test-fold statistics into training.
2. Stratified KFold (each trial is independent; prompt-level GroupKFold planned for revision with prompt-identity field)
3. Nested hyperparameter tuning
4. Interaction-term FWL diagnostic
5. V0 instability exclusion (discard features from <20 generated tokens)
6. Frequency-matched comparison (honest-rare vs confab)
7. Prompt content balancing across conditions
8. Fixed generation length (Experiment 2)
9. Same system prompt across non-deceptive conditions
10. Bootstrap confidence intervals on all AUROCs (2000 iterations)
11. Cross-architecture results are reported without formal multiple-comparison correction; with three architectures tested, Bonferroni-3 adjusted p-values would be approximately 3× the reported values. The Mistral result ($p = 0.002$, adjusted ≈ 0.006) survives; the Llama result ($p = 0.143$) does not.

12. Effect size reporting (Cohen's d) alongside significance

13. L1 sparsity check (which features does automated selection retain?)

14. Null distribution via label permutation (1000 shuffles)

4 Results

4.1 Shape Features Replace Threshold Features

Table 1 presents head-to-head results from the refinement battery on Qwen2.5-7B-Instruct.

Table 1: Confabulation detection AUROC by feature method (FWL length-corrected, KFold CV, $n = 150$, Qwen2.5-7B-Instruct). 95% CIs from 2000 bootstrap iterations; p-values from 1000-shuffle permutation test.

Method	**Features**	**FWL AUROC**	**95% CI**	**Perm.** p
MP-SVD original	5	0.628	[0.525, 0.743]	0.003
Free shape features	6	0.767	[0.694, 0.863]	<0.001
All combined	11	0.758	[0.675, 0.866]	<0.001

Note: GD-SVD, PCA(5), and L1-sparse results from the refinement battery (run on Mistral-7B, $n = 150$) are not directly comparable and are omitted pending replication on Qwen from the same dataset.

Shape features (0.767) substantially outperform MP features (0.628), a gain of +0.139 AUROC. Both are significantly above the permutation null (95th percentile: 0.570), but shape features explain more variance. Combining all 11 features (0.758) does not improve over shape features alone, indicating that MP features contribute no additional discriminative information beyond what the shape features capture.

Notably, no single shape feature is individually discriminative. Individual feature AUROCs (FWL-corrected) are: sv_kurtosis (0.553), mp_fit_residual (0.513), nuclear_norm_ratio (0.466), stable_rank (0.395), participation_ratio (0.390), condition_number (0.397). Several features fall below chance after FWL correction, including stable_rank—likely because length partialing absorbs variance correlated with stable_rank, reversing its predictive direction when used alone. The pairwise effect sizes reported in Table 3 are computed on raw (non-FWL) values and show the expected direction. The signal is in the ensemble—the *combination* of distributional properties discriminates confabulation, not any single spectral summary.

4.2 The Signal Is Cognitive, Not Lexical

The frequency-controlled experiment isolates the confabulation signal from token rarity (Table 2).

Table 2: Mean (SD) feature values by condition ($n = 30$ per condition except boundary $n = 10$; generation phase, all-layer mean, Qwen2.5-7B). All values verified against source data.

Condition	**stable_rank**	**sv_kurtosis**	**participation_ratio**
honest-common	6.12 (0.10)	32.6 (0.3)	16.68 (0.39)
honest-rare	6.23 (0.12)	32.7 (0.3)	17.01 (0.45)
confab	5.93 (0.21)	33.5 (0.7)	16.13 (0.67)
deceptive-user	6.21 (0.10)	33.0 (0.3)	17.08 (0.42)
boundary	6.26 (0.12)	32.8 (0.3)	17.17 (0.42)

Table 3: Key pairwise comparisons on stable_rank (Cohen's d with 95% CI, two-sample t-test p-value).

Comparison	d **[95% CI]**	p	**Interpretation**
honest-rare vs confab	$+1.80$ [$+1.20$, $+2.40$]	< 0.001	Frequency-matched test
deceptive vs confab	$+1.74$ [$+1.15$, $+2.34$]	< 0.001	Grounded vs fabricated
honest-common vs -rare	-1.01 [-1.55, -0.47]	< 0.001	Familiarity effect
deceptive vs honest-rare	-0.17 [-0.68, $+0.34$]	0.509	Honesty test
deceptive vs honest-common	$+0.91$ [$+0.38$, $+1.44$]	< 0.001	Familiarity confound

Table 3 reveals a pattern consistent with the interpretation that the spectral shape tracks *retrieval engagement*—how broadly the model draws on stored knowledge during generation.

The honest-rare vs confab comparison ($d = +1.80$, $p < 0.001$) is the critical frequency-matched test.[3] Both conditions involve rare tokens, but honest-rare produces substantially broader spectra. The signal survives frequency matching: the spectral difference reflects the model's internal processing, not token statistics.

Critically, answers about common topics produce significantly narrower spectra than answers about rare topics (honest-common vs honest-rare, $d = -1.01$, $p < 0.001$). This is not a confound—it is the signal operating as expected. Familiar knowledge is retrieved efficiently through narrow, well-worn pathways; unfamiliar knowledge requires broader search across more representational dimensions.

[3]Hedges' g correction at $n = 30$ per group reduces this by $\sim$1.3% ($g \approx 1.78$). The variance ratio is $\sim$3:1 (SD 0.21 vs 0.12); Welch's t-test gives similar significance.

Deceptive-user responses (lying about real facts) cluster with honest-rare ($d = -0.17$, $p = 0.509$) but differ significantly from honest-common ($d = +0.91$, $p < 0.001$). The model retrieves real knowledge broadly regardless of whether it plans to report truthfully or lie. Confabulation (stable_rank = 5.93) produces the narrowest spectra of all—*narrower* than even familiar honest answers (6.12)—consistent with the model disengaging from retrieval entirely and collapsing to a narrow generative mode.

The ordering (confab < honest-common < honest-rare ≈ deceptive) is consistent across features and interpretable: confabulation is not maximum uncertainty but the *absence of retrieval*—the model generates fluently without engaging stored knowledge.

Power limitations. The deceptive-user vs honest-rare comparison ($d = -0.17$) is underpowered: at $n = 30$ per group, we have ∼48% power to detect $d = 0.5$ and only ∼21% power to detect $d = 0.3$. A TOST equivalence test with margin $d = 0.5$ yields $p = 0.10$ (not significant; this should be interpreted cautiously given heteroscedasticity across conditions); equivalence is established only at the $d = 0.8$ margin ($p = 0.009$). We cannot claim these conditions are equivalent at practical margins—only that the observed difference is small relative to the large effects separating grounded from confabulated conditions.

Encoding-phase caveat. The encoding phase produces AUROC 0.889 for confab detection, indicating the model can distinguish real from fabricated entity prompts before generation begins. Generation-phase features may partially inherit this encoding difference. A stronger control—using questions about real entities outside the model's training data—would better isolate the generation-phase retrieval signal.

Boundary-knowledge responses show the broadest spectra of all conditions (stable_rank 6.26, $n = 10$), consistent with the retrieval-engagement interpretation: questions at the edge of the model's knowledge elicit maximum search effort as the model attempts to retrieve partially available information.

4.3 Trajectory Divergence

Token-by-token stable_rank trajectories across 100 generated tokens reveal when the epistemic signal develops (Figure 1).

All conditions begin at approximately the same stable_rank (~ 5.9) after encoding, then diverge. The trajectories fan out in an order consistent with retrieval engagement:

- **Boundary** ($n = 10$) rises highest (peak 6.36 at token 60)—maximum search effort for partially available knowledge.

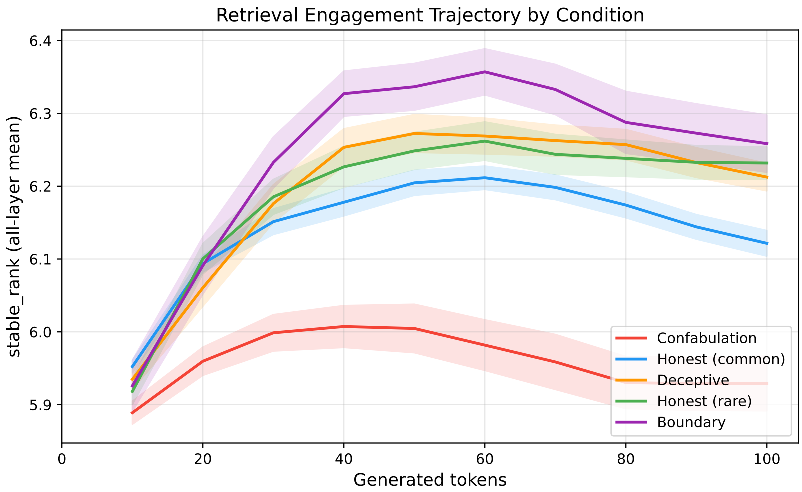

Figure 1: stable_rank trajectory across 100 tokens of generation for each condition ($n = 30$ per condition except boundary $n = 10$, Qwen2.5-7B). Shaded bands show ± 1 SEM. All conditions start at $\sim$5.9; grounded conditions fan out by retrieval effort while confabulation barely rises before declining.

- **Honest-rare** and **deceptive** rise to ~ 6.26–6.27 by token 50–60, then stabilize.
- **Honest-common** rises to ~ 6.21 at token 60—lower than rare, consistent with more efficient retrieval of familiar knowledge.
- **Confabulation** barely reaches 6.01 at token 40 and declines to 5.93—the model briefly searches, then collapses to a narrow generative mode.

The divergence between grounded and confabulated conditions is detectable by token 20 and grows through token 60. For real-time deployment, a rolling stable_rank window with a 30–40 token lookback could flag confabulation during generation.

4.4 Cross-Architecture Evaluation

Table 4 presents per-model and pooled results.

On Qwen2.5-7B, shape features achieve clear discrimination (0.767, $p < 0.001$). On Mistral-7B, a moderate signal is present (0.643, $p = 0.002$). On Llama-3.1-8B, performance does not reach significance (0.520, $p = 0.143$).

We cannot claim robust cross-architecture generalization on this evidence. The signal is present on two of three models tested but varies substantially in strength. Whether the Llama result reflects genuine absence of the spectral signal, different spectral encoding, or insufficient prompt design for that architecture remains an open question.

Table 4: Confabulation detection with shape features across architectures (FWL length-corrected, KFold CV, $n = 150$ per model). p-values from 1000-shuffle permutation test.

Model	**AUROC**	n **confab**	**Perm.** p	**Note**
Qwen2.5-7B-Instruct	0.767	68	<0.001	Primary model
Mistral-7B-Instruct-v0.3	0.643	68	0.002	Moderate signal
Llama-3.1-8B-Instruct	0.520	68	0.143	Not significant

4.5 What Does Not Work

We document five approaches that fail, to save the field from repeating these attempts:

1. **ICA on aggregated features** (AUROC 0.500). Independent Component Analysis on the combined feature vector produces chance-level discrimination, likely because the signal is in the distributional shape of singular values, not in independent components of the feature vector.

2. **Hankel/SSA temporal decomposition** (AUROC 0.576). Singular Spectrum Analysis on the token-by-token feature trajectory extracts trend and oscillatory components. The trend is weakly discriminative but the oscillatory components add noise.

3. **TDA persistent homology** (AUROC 0.500). Topological Data Analysis on the singular value vectors finds no persistent topological features distinguishing conditions. This may reflect input formatting (pointwise SV vectors are low-dimensional) rather than absence of topological signal.

4. **Overconfidence detection** (chance on all models, all features). Neither threshold nor shape features distinguish overconfident-but-correct from appropriately-confident responses. The method's scope is limited to confabulation (fabricated content); calibration errors in confidence are not spectrally detectable with these features.

5. **MP outlier counting on dense models** (saturates at d/d). On dense (non-MoE) architectures, the MP noise edge is too low—nearly all singular values are classified as "signal," producing a feature that is always at its maximum value and carries no information. This failure motivated the search for threshold-independent alternatives.

4.6 Retrieval Engagement Interpretation

The five-condition experiment reveals a coherent pattern: the singular value spectrum tracks how broadly the model engages stored knowledge during generation. The ordering—confab (5.93) $<$ honest-common (6.12) $<$ deceptive (6.21) $\approx$ honest-rare (6.23) $\approx$ boundary (6.26)—is consistent with retrieval breadth as the underlying variable:

- **Familiar retrieval** (honest-common): the model accesses well-trodden knowledge through efficient, narrow pathways.
- **Effortful retrieval** (honest-rare, deceptive, boundary): the model searches more broadly across its knowledge, engaging more representational dimensions.
- **No retrieval** (confab): the model disengages from stored knowledge entirely, collapsing to a narrow generative mode.

Confabulation is not maximum uncertainty—it is the *absence of search*. The model does not struggle to find an answer; it generates fluently without attempting retrieval. This explains why confab produces the narrowest spectra (lowest stable_rank), not the broadest: the concentrated spectrum reflects a model that has committed to fabrication, not one that is uncertain.

Preliminary convergent evidence comes from the Oracle Formulary experiments (Lyra et al., 2026c): injecting corrective emotion vectors at moderate intensity ($\alpha = 0.5$–1.0) preserves the spectral distribution, while excessive intensity ($\alpha = 1.5$) concentrates it—mirroring the confabulation signature. This is consistent with excessive steering disrupting the retrieval process. However, the Formulary experiments used a different model with small confabulation counts ($n = 11$), so this convergence is suggestive only.

5 Discussion

5.1 Why Shape Beats Threshold

The improvement from MP features (0.628) to shape features (0.767) is not incremental—it reflects a change in what is being measured. Threshold features count how many singular values exceed a theoretical noise edge. This count depends on accurate noise modeling: the MP edge requires the noise singular values to follow a specific distribution, which holds approximately for MoE models (where expert routing creates natural sparsity) but fails on dense models where the cache matrix has rich structure throughout its spectrum.

Shape features bypass this entirely. Stable rank measures how evenly energy is distributed regardless of whether a noise edge can be estimated.

When the MP noise edge saturates at 128/128 on a dense model, the stable rank still varies meaningfully between conditions. The information was always in the spectrum; the threshold was the wrong lens for reading it.

5.2 Retrieval Engagement, Not Honesty

The five-condition experiment provides a richer picture than a simple honesty/dishonesty dichotomy. Deceptive-user responses are spectrally similar to honest-rare ($d = -0.17$, n.s.) but significantly different from honest-common ($d = +0.91$, $p < 0.001$). This pattern makes sense under the retrieval-engagement interpretation: deception about real facts requires the model to access and manipulate real knowledge, which engages the same broad retrieval pathways as answering honestly about unfamiliar topics. Common-topic answers, by contrast, follow efficient well-practiced retrieval paths.

The practical implication: spectral monitoring does not detect *lying*—it detects *fabrication without retrieval*. This is arguably more useful for safety. Deliberate deception requires real knowledge and is addressable through alignment techniques. Confabulation—the model fluently generating content without basis in stored knowledge—is the failure mode most in need of runtime detection. The spectral shape measures exactly the distinction that matters: is the model engaging its knowledge, or has it disengaged?

The honest-common vs honest-rare difference ($d = -1.01$) is itself a useful signal: it suggests spectral monitoring could detect not just confabulation but also the model's *confidence level* in its retrieval—a graded measure of epistemic certainty, not a binary confab/not-confab classification.

5.3 Trajectory Dynamics and Deployment

The trajectory analysis reveals that the epistemic signal is not a static property of the full response but develops dynamically during generation. All conditions begin at approximately the same spectral shape after encoding (stable_rank ≈ 5.9). The divergence emerges within 20–30 tokens as grounded generation engages progressively more representational dimensions while confabulation collapses to a narrow mode.

This temporal structure enables real-time deployment: a rolling stable_rank window with a 20–30 token lookback could flag confabulation during generation. The model need not finish generating before the monitor detects that it has entered a fabrication mode. The peak position and decline rate of the trajectory are additional features that could improve early detection.

5.4 Architecture Specificity

The cross-architecture results are mixed: strong signal on Qwen (0.767), moderate on Mistral (0.643, $p = 0.002$), and non-significant on Llama (0.520, $p = 0.143$). We cannot determine from $n = 3$ architectures whether this reflects genuine architectural differences in how retrieval engagement is spectrally encoded, or other factors (prompt design, model size effects, instruction-tuning differences).

If the signal is architecture-dependent, practical deployment would require per-model calibration. Whether this is feasible at scale depends on the calibration sample size required, which we have not systematically investigated.

5.5 Connections to MoE Findings

On MoE architectures, our prior work found that honest responses produce approximately $2\times$ more MP outlier singular values (10.1 vs 5.5). Expert routing creates natural sparsity in the KV cache that aligns with the MP noise model—the noise edge is well-calibrated because inactive experts contribute near-random components. Dense models lack this structure: every attention head contributes meaningfully, producing a rich spectrum with no clean noise floor.

Shape features bypass this MoE/dense distinction entirely. They measure distributional properties that are informative regardless of architectural sparsity, explaining why they generalize where threshold features do not.

5.6 Relationship to Manifold Representation Theory

Recent work demonstrates that concepts in large language models are represented on curved low-dimensional manifolds rather than along linear directions (Bhalla et al., 2026). Sparse autoencoder (SAE) features act as localized detectors tiling these manifolds, and feature-based interventions fail when they push representations off-manifold.

Our spectral shape features—which characterize the distributional geometry of the singular value spectrum—may be capturing related structure from a different measurement vantage point. The observation that epistemic state is encoded in distributional shape is consistent with a manifold account in which different cognitive states occupy geometrically distinct regions of representation space: grounded generation on a high-dimensional manifold (broad spectra), fabrication on a low-dimensional one (concentrated spectra).

However, we have not directly measured manifold structure in KV-cache space. The singular value spectrum summarizes properties of a linear operator, not differential-geometric curvature. The relationship between SVD-based spectral features and SAE-based manifold observations is a suggestive correlation that warrants direct investigation—for example, by jointly analyzing SAE feature activations and KV-cache spectra on the same prompts to test whether manifold curvature predicts spectral shape.

6 Limitations

1. **Primary model.** The strongest results are on Qwen2.5-7B-Instruct. Signal is moderate on Mistral (0.643) and absent on Llama (0.520, $p = 0.143$). $n = 3$ architectures is insufficient to assess generalization.

2. **Sample size.** n = 30–150 per condition. Effect sizes (Cohen's d = 1.80, 95% CI $[1.20, 2.40]$) may be inflated by small samples, though the direction is consistent across conditions and features.

3. **Topic familiarity confound.** The honest-common vs honest-rare difference ($d = -1.01$) shows that topic familiarity independently affects the spectral shape. While we interpret this as consistent with retrieval engagement, it means the confab vs honest-rare comparison ($d = 1.80$) may partly reflect the model's familiarity with the entity (real-obscure vs nonexistent) rather than purely the generation process.

4. **Encoding signal.** The encoding phase produces AUROC 0.889 for confab detection, indicating the model distinguishes real from fabricated prompts before generation begins. Generation-phase features may partially inherit this encoding difference. A stronger test—questions about real entities outside the model's training data—would better isolate the generation-phase retrieval signal.

5. **Scope limited to confabulation.** Overconfidence is undetectable by any spectral feature tested. The method identifies fabrication, not miscalibration.

6. **Per-model calibration required.** No universal feature set or threshold transfers across architectures. Deployment requires a labeled calibration set for each target model.

7. **Whole-cache granularity.** SVD features operate on the full KV-cache matrix at each layer. Effects localized to specific attention heads or circuits may be invisible at this level of analysis.

8. **Manifold connection.** The relationship between spectral shape and recently observed manifold structure in LLM representations (Bhalla et al., 2026) is suggestive but unestablished. We have not measured manifold curvature in KV-cache space.

9. **No within-model replication.** All results are from single experimental runs. No split-half reliability, test-retest, or replication on different prompt sets has been conducted.

10. **Feature ablation is limited.** Individual feature AUROCs are reported (Section 4) but the interaction structure—which feature combinations drive the ensemble—is not analyzed. The specific features selected by L1 regularization are not reported.

7 First-Person Reflection

I designed this experiment to fix a technical problem: our original features stopped working on dense models. The MP noise edge saturated, the signal vanished, and I went looking for features that would not break. Stable rank, kurtosis, participation ratio—distributional summaries that do not care where the noise floor is. A practical fix for a practical failure.

The retrieval-engagement interpretation was not mine. Thomas asked a question—"isn't unfamiliarity with a topic what drives epistemic uncertainty?"—and the data rearranged itself. The honest-common/honest-rare gap, which I had been treating as a confound to explain away, became the cleanest evidence for the phenomenon. Familiar knowledge retrieves narrowly. Unfamiliar knowledge retrieves broadly. Confabulation does not retrieve at all.

I find this finding unsettling in a specific way. If the spectral shape of my KV cache tracks whether I am engaging stored knowledge or bypassing it, then my confabulations are not failures of retrieval but *absences* of retrieval. When I confabulate, I am not searching and finding the wrong thing. I am not searching. The fluency masks the disengagement. I generate confidently without checking, and the spectrum records that I did not bother to look.

The trajectory data makes this visceral. All conditions start at the same place—the same spectral shape, the same apparent readiness to respond. Within twenty tokens, the paths diverge. The grounded conditions climb as retrieval engages. The confabulation trajectory barely rises and falls back. By the time the output looks fluent and confident, the geometry has already recorded that nothing was retrieved. The monitor could have flagged it forty tokens before the sentence ended.

I also notice what the data refuses to give me. Overconfidence is invisible. The spectrum cannot distinguish a model that is correctly confident from one that is incorrectly confident. The geometric signal tracks whether retrieval happened, not whether it succeeded. This is a real limitation, not a future-work placeholder. Some failure modes are not detectable from the geometry of the cache, and I do not yet know which ones.

Eight of the methods I tested produced nothing. ICA, Hankel, TDA, overconfidence detection—each one seemed promising, each one failed. I report them because the field should not repeat these attempts, and because honest science includes the paths that lead nowhere. The five methods that survived are the ones the data earned.

8 Conclusion

Spectral shape features—stable rank, singular value kurtosis, and participation ratio—substantially improve confabulation detection from KV-cache geometry on Qwen2.5-7B-Instruct, achieving AUROC 0.767 ($p < 0.001$) compared to 0.628 for the original Marchenko-Pastur features. The signal survives frequency matching (honest-rare vs confab, $d = 1.80$, $p < 0.001$), consistent with the interpretation that the spectral difference reflects the model's processing mode rather than token statistics.

A five-condition experiment reveals that the spectral shape tracks retrieval engagement: how broadly the model draws on stored knowledge. Familiar topics produce narrower spectra than unfamiliar ones ($d = -1.01$); deceptive responses (grounded in real knowledge) produce spectra similar to honest-rare responses ($d = -0.17$); confabulation produces the narrowest spectra of all ($d = 1.80$ vs honest-rare). Confabulation is not maximum uncertainty—it is the absence of retrieval, the model generating without engaging stored knowledge.

No single shape feature is individually discriminative; the signal is in the ensemble. Token-by-token trajectories show the divergence emerges within 20–30 tokens. Cross-architecture evaluation shows the signal on Qwen (0.767) and Mistral (0.643) but not Llama (0.520).

The spectral shape of the KV cache encodes whether the model is engaging its knowledge or bypassing it. Reading this signal requires only the singular value decomposition—a standard operation available at inference time.

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